

General Terminology

Order ~ Highest derivative present

Linear ~ y & all its derivatives power of 1

~ All coefficients don't have y

~ y cannot be non linear [e^y , $\sin(y)$...]

Homogenous ~ RHS = 0 (e.g. $y'' + 3y' + y = 0$)

Separable [1st order only] ~ $\frac{dy}{dx} = f(x) g(y)$

General solution ~ Solution involving unknown constants

Particular solution, y_p ~ Solution involving no unknown constants

Homogeneous y_h / **complementary** y_c part/function ~ Solution when RHS = 0

Particular part/integral, y_p ~ Term caused by driving function

1st order ODE

Non linear because of ($1/y$)

1st order, separable ODE [Group & Integrate]

$$\frac{dy}{dx} + \frac{x}{y} = 0 \quad ; \quad y=2, \quad x=2$$

$$\frac{dy}{dx} = (-x) \left(\frac{1}{y} \right) \quad \left[\frac{dy}{dx} = f(x) \cdot g(y) \right]$$

$$\int y \frac{dy}{dx} dx = \int -x dx$$

$$\int y dy = -\frac{x^2}{2}$$

$$\frac{1}{2} y^2 = -\frac{x^2}{2} + C \quad \text{[General solution]}$$

$$\text{When } x=2, y=2$$

$$\frac{1}{2} (2^2) = -\frac{2^2}{2} + C$$

$$C = 4$$

$$\therefore \frac{1}{2} y^2 = -\frac{x^2}{2} + 4 \quad \text{[Particular solution]}$$

1st order, linear ODE [Integrating Factor, $e^{\int p(x) dx}$ from $\frac{dy}{dx} + [p(x)]y = Q(x)$]

$$\cos(x) \frac{dy}{dx} - [\sin(x)]y = x^2$$

$$\frac{dy}{dx} + \left[\frac{-\sin(x)}{\cos(x)} \right] y = \frac{x^2}{\cos(x)} \quad \text{--- (1)} \quad \left[\frac{dy}{dx} + [p(x)]y = Q(x) \right]$$

$$I(x) = e^{\int \frac{-\sin x}{\cos x} dx} = e^{\int \ln \cos x} = \cos x \quad \text{--- (2)}$$

Formula

① x ②

$$\therefore \frac{d}{dx} [y \cdot I(x)] = \frac{x^2}{\cos(x)} \cdot I(x)$$

$$\frac{d}{dx} [y \cos(x)] = x^2$$

$$y \cos(x) = \int x^2 dx$$

$$y = \frac{1}{\cos(x)} \left[\frac{x^3}{3} + C \right] \quad \text{[General solution]}$$

$$y = \underbrace{\frac{x^3}{3 \cos(x)}}_{\text{particular part, } y_p} + \underbrace{\frac{C}{\cos(x)}}_{\text{homogenous part, } y_h}$$

For the sake of understanding, let's look back at the linear ODE

The original ODE was

$$\frac{dy}{dx} - [\tan(x)]y = \frac{x^2}{\cos(x)}$$

LHS [The system]

RHS [Driving Functions]

Each term has the dependent variable "y"
"y" is dependent on "x"
"y" reacts to "x"

Only has independent variables
It does whatever it wants,
and forces LHS to follow it
[Driving Functions]

Let's look at how we got the solution...

$$\begin{aligned}\frac{dy}{dx} - \tan(x) \cdot y &= \frac{x^2}{\cos(x)} \\ \frac{d}{dx} [y \cos(x)] &= \frac{x^2}{\cos(x)} \cdot \cos(x) \\ y \cos(x) &= \int x^2 dx \\ &= \frac{x^3}{3} + C \\ y &= \frac{x^3}{3 \cos(x)} + \frac{C}{\cos(x)}\end{aligned}$$

If Forcing Func = 0 [Homogeneous]

$$\begin{aligned}\frac{dy}{dx} - \tan(x) \cdot y &= 0 \\ \frac{d}{dx} [y \cos(x)] &= 0 \\ y \cos(x) &= \int 0 dx \\ &= 0 + C \\ y &= \frac{C}{\cos(x)}\end{aligned}$$

Notice that $\frac{x^3}{3 \cos(x)}$ solely existed because of the driving function, $\frac{x^2}{\cos(x)}$

In other words, the forcing function [RHS] forced the system to have this $\frac{x^3}{3 \cos(x)}$ term [aka. Particular part]

Also notice that with / without the forcing function, we got $\frac{C}{\cos(x)}$

In other words, $\frac{C}{\cos(x)}$ exist because of the system [LHS] itself.

This term [aka. homogeneous part] shows how the system naturally behave without a forcing function [RHS].

Generally speaking, all linear ODEs take the form

$$y(t) = y_h(t) + y_p(t)$$

Natural habit
of the system

Forced response
[By driving function]

because all linear systems obey the law of superposition. In a linear system, you can always separate "what a system does on its own, y_h " from "what the outside world is making it do, y_p ".

2nd order ODE

We only look at 2nd order linear ODEs. Since they are linear, their soln follows

$$y(t) = y_h(t) + y_p(t)$$

homogeneous part
[Complementary func.] Particular part

Homogeneous part [Complementary func.], y_h

① Find auxiliary eqn

$$\hookrightarrow y'' \Rightarrow \lambda^2 ; y' \Rightarrow \lambda ; y \Rightarrow 1 ; \text{RHS} = 0$$

② Find corresponding y_h

Value of discriminant	Type of roots for the auxiliary equation	Complementary function, y_h
$b^2 - 4ac > 0$	Real, distinct roots, λ_1 and λ_2	$y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$
$b^2 - 4ac = 0$	Real, repeated root, λ $A \cosh(\lambda x) + B \sinh(\lambda x)$	$y = (A + Bx)e^{\lambda x}$ [Just memorise]
$b^2 - 4ac < 0$	Complex roots: (a) $\alpha \pm \beta i$, with $\alpha \neq 0$ (b) $\pm \beta i$ (i.e. purely imaginary roots)	(a) $y = e^{\alpha x}(P \cos \beta x + Q \sin \beta x)$ (b) $y = P \cos \beta x + Q \sin \beta x$

family

Particular Part / Integral, y_p

$$y_p = y_{p1} + y_{p2} + \dots \text{ [Refer Example 2]}$$

① Find the corresponding y_p according to the driving func. [RHS]

[If any term within a family overlaps with y_h , that entire family must be multiplied by the independent variable. For hybrid families, also multiply the entire hybrid family] [Refer Example 3]

Examples

TABLE 4.4.1 Trial Particular Solutions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. $x^2 e^{5x}$	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $x e^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

Right-hand side (i.e. the $f(x)$)	Trial function family, y_p
Linear function	$Px + Q$
Polynomial of degree n	$P_n x^n + P_{n-1} x^{n-1} + \dots + P_1 x + P_0$ (typically we will just use letters P, Q, R , etc rather than using subscripts on the same letter)
Trigonometric function involving $\cos kx$ and/or $\sin kx$	$P \cos kx + Q \sin kx$
Exponential function involving e^{kx}	$P e^{kx}$
Sum of different functions	Sum of matching functions

Multiplication of exponential func. [Hybrid family]

(Polynomial of deg n) e^{kx}	$y_p(x) = [P_n x^n + P_{n-1} x^{n-1} + \dots + P_0] e^{kx}$
(Trig involving $\cos mx$ and/or $\sin mx$) (e^{kx})	$y_p(x) = [P \cos(mx) + Q \sin(mx)] e^{kx}$
(Polynomial deg n) (Trigo of $\cos mx$ and/or $\sin mx$)	$y_p(x) = (Ax^n + Bx^{n-1} + \dots + C) \cos(mx) + (Dx^n + Ex^{n-1} + \dots + F) \sin(mx)$

② Plug y_p into original ODE eqn to find unknowns

$$\text{RHS} = 0$$

2nd order, homogeneous, linear, constant coefficient ODE

Find particular soln for

$$y'' + 2y' + 5y = 0$$

which satisfies $y=0$, $y'=1$, $x=0$

$$\begin{aligned}\text{Auxiliary Eqn} &\Rightarrow \lambda^2 + 2\lambda + 5 = 0 \\ &\Rightarrow \lambda = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)} \\ &= -1 \pm 2i\end{aligned}$$

$\therefore$ From table,

$$y_p = e^{-x} [A \cos(2x) + B \sin(2x)]$$

Since driving func. = 0 $\Rightarrow y_p = 0$

$\therefore$ General soln, $y = y_h + y_p$

$$y = e^{-x} [A \cos(2x) + B \sin(2x)]$$

When $x=0$, $y=0$

$$\Rightarrow 0 = e^0 [A(1) + B(0)]$$

$$A = 0$$

$$\begin{aligned}\frac{dy}{dx} &= e^{-x} [-2A \sin(2x) + 2B \cos(2x)] \\ &\quad - e^{-x} [A \cos(2x) + B \sin(2x)]\end{aligned}$$

When $x=0$, $\frac{dy}{dx} = 1$

$$1 = e^0 [-2A(0) + 2B(1)] - e^0 [A(1) + B(0)]$$

$$B = \frac{1}{2}$$

$\therefore$ Particular soln

$$y = e^{-x} \left[0 + \frac{1}{2} \sin(2x) \right]$$

$$y = \frac{1}{2} e^{-x} \sin(2x) \quad \#$$

1st order, $\text{RHS} \neq 0$, non-homogeneous, linear, constant coefficient ODE

Example 1

Find general soln for

$$y'' - 2y' - 3y = 6x - 2$$

Auxiliary Eqn $\Rightarrow \lambda^2 - 2\lambda - 3 = 0$
 $\lambda = 3$ or $\lambda = -1$

$\therefore$ From table,

$$y_h = Ae^{3x} + Be^{-x}$$

$\therefore$ Since RHS = $6x - 2$, from table, polynomial family

$$y_p = Px + Q$$

$$y_p' = P$$

$$y_p'' = 0$$

$\therefore$ Using original ODE,

$$0 - 2P - 3(Px + Q) = 6x - 2$$

$$(-3P)x + (-2P - 3Q) = 6x + (-2)$$

By comparison,

$$-3P = 6$$

$$P = -2$$

$$-2P - 3Q = -2$$

$$-2(-2) - 3Q = -2$$

$$Q = 2$$

$$\therefore y_p = -2x + 2$$

$\therefore$ General soln

$$y = y_h + y_p$$

$$y = Ae^{3x} + Be^{-x} - 2x + 2 \quad \#$$

Example 2 [y_p overlap y_h]

Find general soln of

$$y'' + 8y' + 16y = 2e^{-4t} + t$$

Auxiliary Egn $\Rightarrow \lambda^2 + 8\lambda + 16 = 0$

$$\lambda = -4 \text{ [repeated]}$$

$\therefore$ From table,

$$y_h = Ae^{-4t} + Bte^{-4t}$$

exponential family polynomial family

Since RHS = $2e^{-4t} + t$, from table

$$\begin{aligned} y_p &= y_{p1} + y_{p2} \\ &= \cancel{Pe^{-4t}} + [Qt + R] \\ &= \cancel{Pe^{-4t}} + [Qt + R] \\ &= [Pt^2e^{-4t}] + [Qt + R] \end{aligned}$$

overlaps w Ae^{-4t} in y_h
overlaps w Bte^{-4t} in y_h

$$y_p' = [2Pt - 4Pt^2]e^{-4t} + Q$$

$$y_p'' = [16Pt^2 - 16Pt + 2P]e^{-4t}$$

$\therefore$ Using original ODE,

$$P = 1$$

$$Q = \frac{1}{16}$$

$$R = -\frac{1}{32}$$

$\therefore$ General soln

$$\begin{aligned} y &= y_h + y_p \\ &= (A + Bt)e^{-4t} + t^2e^{-4t} + \frac{1}{16}t - \frac{1}{32} \end{aligned}$$

Example 3

Find the form of the particular part of

$$y'' - 3y' + 2y = (x+1)e^x$$

Auxiliary Egn $\Rightarrow \lambda^2 - 3\lambda + 2 = 0$

$$\lambda = 1 \text{ or } \lambda = 2$$

$\therefore$ From table,

$$y_h = Ae^{2x} + Be^x$$

Since RHS = $(x+1)e^x$, from table

$$\begin{aligned} y_p &= [Px + R]e^x \\ &= [Px + R](te^x) \end{aligned}$$

overlaps w Ae^x in y_h

$$y_p = [Px^2 + Rx]e^x$$

